

MARAGHEH, Iran, Islamic Republic of

WMO: 407130

Lat: 37.3477N

Lon: 46.1461E

Elev: 4849

StdP: 12.30

Time Zone: 3.50 (IRN)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	12.5	17.2	3.3	7.8	16.8	7.2	9.5	24.6	21.4	35.0	18.8	38.4	4.4	80	0.541

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	22.0	98.2	63.2	95.0	62.2	92.6	61.7	65.9	91.3	64.6	89.8	63.4	88.2	7.7	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
56.2	80.5	79.9	54.5	75.7	78.5	53.2	72.1	76.9	33.4	91.7	32.4	89.6	31.4	87.9	71.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
23.4	20.8	18.3	8.5	102.0	7.0	2.4	3.5	103.8	-0.6	105.2	-4.6	106.6	-9.7	108.3
			6.9	69.1	6.3	1.4	2.3	70.1	-1.4	70.9	-4.9	71.7	-9.5	72.7
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	57.2	32.9	37.1	45.6	54.1	64.2	74.9	81.3	81.2	72.2	59.6	45.5	36.3	(6)
(7)		DBStd	18.15	8.02	7.32	7.37	6.57	5.22	4.72	4.60	4.26	5.59	6.46	6.77	8.04	(7)
(8)		HDD50	1699	531	363	175	31	0	0	0	0	7	165	427	(8)	
(9)		HDD65	4467	996	782	602	331	80	1	0	10	189	586	890	(9)	
(10)		CDD50	4317	0	1	38	154	439	746	969	967	667	305	29	2	(10)
(11)		CDD65	1611	0	0	0	5	54	297	504	502	227	22	0	0	(11)
(12)		CDH74	17591	0	0	2	55	503	3079	5888	5765	2109	190	0	0	(12)
(13)		CDH80	8279	0	0	0	10	80	1302	3082	2994	788	23	0	0	(13)
(14)	Wind	WSAvg	7.9	5.7	6.7	8.0	8.2	8.4	9.4	10.8	10.5	8.6	7.4	6.2	5.3	(14)
(15)	Precipitation	PrecAvg	11.4	1.1	1.0	1.6	2.3	1.3	0.2	0.1	0.0	0.2	0.9	1.4	1.1	(15)
(16)		PrecMax	20.4	2.4	3.4	3.5	4.7	3.7	1.6	0.6	0.2	1.5	4.4	3.6	3.7	(16)
(17)		PrecMin	7.1	0.3	0.0	0.4	0.3	0.2	0.0	0.0	0.0	0.0	0.0	0.0	0.1	(17)
(18)		PrecStd	3.2	0.6	0.7	0.7	1.2	0.9	0.4	0.2	0.0	0.4	1.1	0.9	0.8	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	53.6	58.9	72.1	80.3	85.8	95.4	102.2	100.7	94.3	82.6	67.7	58.9	(19)
MCWB			41.0	44.0	49.5	54.5	58.4	63.0	65.2	63.4	60.6	55.9	49.6	43.6	(20)	
2%		DB	49.9	54.4	66.4	73.6	82.2	92.4	98.7	98.0	91.0	79.0	62.9	53.5	(21)	
		MCWB	39.9	42.1	47.5	52.1	57.2	61.2	64.0	62.5	59.8	54.6	47.0	42.4	(22)	
5%		DB	46.7	50.9	61.9	69.7	79.2	89.8	95.3	95.1	87.7	75.4	59.1	49.9	(23)	
		MCWB	38.9	40.3	45.3	50.9	56.1	60.2	62.8	61.8	58.7	53.5	46.1	41.1	(24)	
10%		DB	43.9	47.9	57.9	66.1	76.6	87.4	92.8	92.8	84.6	71.8	55.8	46.6	(25)	
		MCWB	37.2	39.2	43.8	49.0	55.3	59.6	62.5	61.3	57.4	52.2	44.9	39.2	(26)	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	43.9	46.3	51.0	56.3	62.4	65.5	68.2	67.2	63.2	58.0	52.0	46.0	(27)
MCDWB			50.4	55.9	66.4	72.4	79.4	91.4	93.8	91.9	86.6	77.6	61.6	53.9	(28)	
2%		WB	41.4	43.6	48.5	54.3	59.7	63.5	66.4	65.5	61.6	56.0	50.0	44.0	(29)	
		MCDWB	47.4	51.6	62.9	68.6	76.9	88.3	92.3	90.3	85.0	73.9	57.4	51.3	(30)	
5%		WB	39.4	41.5	46.5	52.5	57.9	62.0	65.2	64.2	60.3	54.7	48.3	42.0	(31)	
		MCDWB	45.5	48.6	59.4	65.3	75.5	86.5	90.8	89.4	83.5	71.5	55.9	48.3	(32)	
10%		WB	37.4	39.7	44.8	50.9	56.4	60.6	64.0	63.0	59.0	53.4	46.4	40.0	(33)	
		MCDWB	43.1	46.5	56.3	62.7	73.2	84.1	89.0	88.2	82.1	68.7	54.6	45.7	(34)	
(35)	Mean Daily Temperature Range	5% DB	MDBR	14.0	15.8	17.6	18.8	21.1	23.1	22.0	22.5	22.6	19.7	16.6	13.8	(35)
MCDBR			16.7	19.9	22.6	23.0	24.0	24.9	23.8	23.9	24.6	23.9	20.9	18.1	(36)	
MCWBR			10.0	11.1	10.7	10.2	10.1	10.4	9.4	9.7	10.5	10.7	11.2	10.6	(37)	
5% WB		MCDBR	14.9	16.6	19.9	19.7	22.1	23.3	22.1	22.5	22.7	21.5	16.0	15.5	(38)	
		MCWBR	9.5	9.9	10.2	9.5	10.0	10.1	9.2	9.5	10.2	10.0	8.8	9.8	(39)	
(40)	Clear-Sky Solar Irradiance	taub	0.310	0.345	0.397	0.466	0.457	0.420	0.454	0.421	0.372	0.386	0.330	0.306	(40)	
(41)		taud	2.319	2.218	2.084	1.927	1.989	2.125	2.039	2.132	2.260	2.163	2.320	2.360	(41)	
(42)		Ebn at Noon	275	278	272	259	264	273	263	269	278	259	265	269	(42)	
(43)		Edh at Noon	30	38	48	59	56	49	53	48	39	39	30	27	(43)	
(44)	All-Sky Solar Radiation	RadAvg	804	1100	1375	1716	2139	2561	2475	2269	1871	1283	864	675	(44)	
(45)		RadStd	63	108	119	150	121	121	67	71	98	93	109	89	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days				
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
(46)	Station Only	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(46)
(47)	Regional (8 neighbors)	N/A	N/A	N/A	+1.94	N/A	N/A	N/A	N/A	N/A	N/A	(47)

Nomenclature: See separate page